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A novel reactor configuration for continuous virus inactivation

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ABSTRACT

A novel coiled flow inversion reactor (CFIR) has recently been utilized for continuously carrying out various downstream unit operations that require reaction and mixing like refolding, PEGylation and precipitation. In this study, we demonstrate utility of the CFIR for continuous virus inactivation. Low pH virus inactivation for a mAb therapeutic was performed in continuous mode using this reactor and its performance was compared to a parallel batch setup. It was found that both batch and continuous processes yield about equal level of virus inactivation with comparable logarithmic reduction value (LRV) for XMuLV of 4.32 ± 0.26 in batch versus 4.12 ± 0.08 in continuous over a period of 55 min. The data clearly demonstrates that the reactor configuration can be used in a continuous train for virus inactivation without any reduction in inactivation efficiency or inactivation kinetics. Integration of the reactor with the continuous downstream train has also been evaluated by highlighting different possible configurations. Two different case studies depicting the integration of the reactor to a continuous Protein-A capture chromatography under different set of operational conditions have been discussed. In all cases, consistent process performance and product quality attributes were obtained. The proposed reactor offers a suitable configuration for continuous viral inactivation for mAb continuous processing platforms.

1. Introduction

Due to the increased demand and competition in the recent years, there is a constant pressure on manufacturers to reduce the prices of biopharmaceutical drugs [1]. Shifting the mode of manufacturing for biopharmaceuticals from batch to continuous has emerged as a potential solution to this predicament. It has been shown that continuous processing yields a significant reduction in the manufacturing costs [2,3] by increasing productivity, decreasing working volume, and increasing utilization of equipment's and column [4–6].

Today, monoclonal antibodies (mAbs) have emerged as one of the major product of all leading biopharmaceutical manufacturers. More than 50 % of all new approvals in the last 5 years have been of mAb therapeutics [7]. Typical mAb manufacturing occurs in mammalian cell lines like the Chinese Hamster Ovary (CHO) cells, which are inherently susceptible to viral contamination. This necessitates the presence of at least two orthogonal viral clearance steps in the mAb manufacturing process, namely virus filtration and low pH virus inactivation [8]. Additionally, process steps like Protein-A chromatography, anion exchange chromatography and precipitation offer viral clearance together with product purification [3,8,9].

Low pH viral inactivation in mAb manufacturing process is typically performed after Protein A capture chromatography. Since, protein elutes at a lower pH from the Protein A chromatography, it is convenient to follow it up with viral inactivation [10]. This low pH incubation step is targeted at inactivation of enveloped C-type retroviruses [8] (like Murine Leukemia Virus and Lentivirus). Viral inactivation is typically performed at a pH < 4, by incubating the sample at this pH value for 30−120 min [10–12]. Selection of the inactivation condition depends on the viral log reduction kinetics and the stability profile of the mAb molecule under consideration [8]. Longer incubation time at low pH has been shown to result in product loss due to aggregation [13,14]. During low pH viral inactivation in a batch process, the Protein A elute is collected in a vessel, where its pH is decreased to the desired value by addition of an acid under constant stirring. The sample is then immediately transferred to a second vessel where it is incubated for suitable

Abbreviations: mAb, monoclonal antibody; CFIR, coiled flow inversion reactor; LRV, logarithmic reduction value; X-MuLV, xenotropic murine leukemia virus; LVP, large volume plating; HPLC, high performance liquid chromatography; RTD, residence time distribution; HMWI, high molecular weight impurities; GLP, good lab practice.

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time before forward processing [8]. While the batch viral inactivation process is well understood and robust, it is performed manually and offers a lot of challenges for its conversion to continuous mode [15]. In many cases, the static nature of the viral inactivation process presents a significant bottleneck to a user wishing to achieve end-to-end continuous downstream processing [16]. Moreover, viral safety assessment strategies for continuous downstream processing steps is a topic still under discussion and a clear set of regulatory guidelines are awaited [17].

In recent years, a number of technology solutions have been created by researchers to enable continuous viral inactivation. Novel tubular reactor designs [12,16,18], automated multivessel system [15] and column based strategies [19,20] are the most widely explored alternatives. All these strategies offer promising solutions but with limited success. The column based inactivation strategies either utilizes an existing chromatographic step or adds an additional chromatographic step to achieve the incubation time, thereby adding complexity to the process and reduce its productivity [15]. The automated multivessel system from Pall Life Sciences is the first commercially available system. However, the system runs two virus inactivation vessels in staggered mode thus making the system semicontinuous and resulting in higher capital cost [21]. While various novel tubular reactor designs like the Jig in Box (JIB) [16] by Boehringer Ingelheim and the viral inactivation system [18] by Millipore appear to be promising, there are a number of technical challenges associated with tubular reactor that need to be solved, such as the lack of homogeneity in the material (in terms of protein concentration, pH and conductivity) entering and leaving the reactor, discontinuity in the stream entering the reactor and pulsation in associated pumps [15].

Recently, a novel coiled flow inversion reactor [22] has been utilized for performing biotech unit operations that require reaction and mixing. These include refolding [23], precipitation [24] and PEGylation [25]. The reactor configuration is a modified coiled flow reactor with flow inversion at equidistant right angle bents. This flow inversion has been shown to result in more efficient cross sectional mixing with narrow residence time distribution in the laminar flow regime [26]. This configuration has been shown to offer superior control over critical operational conditions like pH [24] and residence time [25]. Researchers in recent publications have also discussed the potential of utilizing the coiled flow inverter (CFI) for virus inactivation [12]. However, these publications only offers theoretical calculations of viral reduction, with no actual viral clearance study that would validate CFI performance [12].

In this study, we have designed a continuous viral inactivation reactor based on the CFIR geometry [22–24]. The reactor configuration consists of two dynamic inline mixers with the CFI present between them. These dynamic inline mixers provide vigorous mixing required for efficient inline acidification and neutralization. Viral clearance studies were performed both in the proposed CFIR as well as the traditional batch operation. It has been demonstrated that viral inactivation capability is not compromised in the proposed setup. Further, integration of the CFIR with the continuous downstream processing train has also been examined.

2. Materials and methods

2.1. Protein feed and reagents

The monoclonal antibody sample used in this study (mAb A) was obtained from a leading Indian biopharma company (confidentiality agreement prohibits disclosure of company identity). The mAb A sample was provided as harvest with titer of 1.75 mg/mL. This harvest sample was purified using continuous Protein A chromatography (MabSelect™ Sure™) based on the protocol given in section 2.6. The obtained Protein A elute was then neutralized with 2 M tris solution. Neutralized Protein A elute was present in 100 mM glycine buffer and had protein concentration of around 7 mg/mL and pH 7. mAb A samples were stored at 4 °C and had a molecular weight of 150 KDa and an isoelectric point of 8.2.

The virus (X-MuLV) were obtained from ATCC (Catalog No- 1447). McCoy's 5 A media was brought from Gibco and cell line used was PG-4 indicator cell line, obtained from ATCC (Catalog No- CRL-2032).

Tris base, Glycine, sodium chloride, sodium dihydrogenphosphate, sodium hydroxide, disodium hydrogen phosphate, hydrochloric acid were obtained from Merck, India. Various working and stock solutions were prepared using these reagents. pH strips were obtained from Hi Media. MabSelect™ Sure™resin was procured from GE Life Sciences.

2.2. Protocol for the viral inactivation assessment

Virus inactivation studies were performed at the virus testing facility of Syngene International Limited, Bangalore, India. Syngene has a state-of-the-art GLP certified biosafety level-2 laboratory for carrying out viral testing and clearance studies. mAb sample was transported to Syngene via temperature controlled shipping (2–8 °C) and was retested before initiating the study (using protocols given in section 2.7). No significant change was observed in the product quality attributes (data not shown). Inactivation studies were initiated by spiking the mAb sample with the enveloped virus Xenotrophic Murine Leukemia Virus (XMuLV). Spiked sample was then acidified to an appropriate target low pH value. The virus stock solution used in the study was initially sonicated to remove virus aggregates and filtered (0.22 μm) prior to spike and all test samples were handled aseptically to avoid the introduction of contamination during assay performance.

Effectiveness of low pH inactivation was calculated by reduction in virus titer. Moreover, kinetics of the inactivation process was estimated by calculating the virus titer of different time samples. Before initiating the viral inactivation experiment, cytotoxicity and interference assays were performed for the protein sample in order to estimate the appropriate sample dilution to be used. A representative sample of mAb Protein A Elute was used for the study.

2.2.1. Cell-based cytotoxicity evaluation

The PG-4 cell line was exposed to various dilutions of the protein (mAb A). The lowest dilution of protein that had no effect on the morphology of cells was determined in order to use in the interference test. The mAb A sample was mock-spiked with 5% (v/v) McCoy's 5A (serum-free medium for PG-4 cell line), the pH of the sample was confirmed to pH 6.50–7.50 and 0.22 μm filtered. The sample was tested for cytotoxicity on PG-4 cells in duplicate wells of a 6 well plate at undiluted (full strength), 2-fold, 5-fold, 10-fold and 20-fold dilutions using appropriate cell culture media as the diluent (McCoy's 5A medium). For cell control/negative control, only McCoy's 5A medium was used for PG-4 cells. The samples and controls were inoculated on to respective cell lines, incubated in the CO_2 incubator and observed on the 7^{th} day.

2.2.2. Cell-based interference evaluation

The pH of mAb sample was confirmed to pH 6.50-7.50 and filtered through a 0.22 μm filter. Samples at nontoxic dilution was used for the interference study. The sample was prepared with McCoy's 5A medium. Dilutions of samples varied as per the cytotoxicity results. Nontoxic and subsequent dilution of the low pH viral inactivation process step samples were spiked with 5% (v/v) of the X-MuLV stock solution. Serial dilutions (10-fold) of the virus spiked samples were performed using nontoxic diluent. In addition, an aliquot of serum-free medium (McCoy's 5A) was spiked with 5% (v/v) of the X-MuLV virus stock solution. Further, serial dilutions (10-fold) of the virus stock were performed using McCoy's 5A media as the diluent. This sample served as the positive control for the interference study, McCoy's 5A media was used as negative control. The virus spiked sample dilutions and controls were tested in respective assays and assayed in triplicate wells per dilution in case of plaque assay (for X-MuLV). The inoculated plates were observed till 7 days.

2.2.3. Quantification of virus titer by plaque assay

Virus titers were determined using the standard plaque assay method and were expressed as PFU/mL [27,28]. A portion of each test and control sample was diluted in serum-free medium (McCoy's 5A to the dilution factor 10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6} and 10^{-7}). Each dilution was then assayed by titration. Large Volume Plating (LVP) was also performed for some test samples. The inoculated plates were observed for 7 days, stained and titer values were calculated.

Titer values calculated in the plaque formation assay were calculated as follows:

$$\frac{PFU}{mL} = \frac{1}{D} * \frac{1}{V} * N \tag{1}$$

Where, D is dilution, V is inoculated volume, and N is the average number of plaques formed at that dilution.

The reduction factor for the process step represents the logarithm of the ratio of the virus load at the beginning of the first process clearance step and at the end of the process clearance step. The virus reduction factor for each of the removal step in batch and continuous mode was calculated as followed:

$$\mathrm{RF} = \log\left(\frac{C_1 V_1}{C_2 V_2}\right) \tag{2}$$

Where, RF is the reduction factor, C1 is the concentration of the virus in the starting material, V1 is the volume of the starting material, C2 is the concentration of the virus in the post-processing material, and V2 is the volume of the post-processing material.

2.3. Viral inactivation study for the batch process

Viral inactivation study for the batch process was performed in a 250 mL bottle with 100 mL of mAb sample maintained under constant stirring on a magnetic stirrer at room temperature. The study was initiated by spiking the sample with XMuLV virus culture to a final virus concentration of 5% (v/v). Post virus spiking, two samples (5 mL each) were withdrawn from the bottle- processing control and the 0 min time point (load) sample. Next, the virus spiked sample was acidified by addition of titrated volume of 0.2 M HCl, under constant stirring (200 rpm), so as to reduce the pH of the sample to 3.5 ± 0.2. The sample was then incubated for a time period of 55 min with sampling at a number of time points- 5 min, 12 min, 25 min and 55 min. A 5 mL sample was withdrawn at each sampling point except the final time point (55 min) where a large sample volume of 20 mL was withdrawn. Collected samples were quenched by adjusting the pH of the sample to 7 ± 0.5 by addition of 2 M Tris base. In order to immediately quench the collected samples, the sampling containers were prefilled with calculated value of Tris base. Two runs (duplicates) were performed using this protocol for viral inactivation assessment.

2.4. Continuous viral inactivation reactor

Virus inactivation study for the continuous process was performed in the continuous viral inactivation reactor (Fig. 1). The reactor configuration consists of two dynamic inline mixers with the CFI present between them (Fig. 1). Two inline dynamic mixing units were placed before and after the CFI for vigorous mixing [23] resulting in efficient acidification and neutralization. The inline dynamic mixing unit consists of a cylindrical PTFE section (length = 28 mm, IDXOD = 1.1 mm X 1.3 mm) with 4 magnetic beads inside it (length = 10 mm, diameter = 3 mm). The unit was stirred using magnetic stirrer. CFI construction (tube diameter and coil diameter) is based on a set of design criteria- Dean number, pitch distance, bend angle and number of turns per coil. In order to achieve the narrowest residence time distribution (RTD) in the CFIR, the boundary conditions for these design parameters, laid down in previous publications [23,24] have to be followed. For the viral inactivation study a new CFIR was therefore built to provide optimal performance (narrow RTD) at the required flow rate. The CFIR was constructed by coiling 1.6 mm ID tubing (Masterflex, L/S 14, Tygon®) around PVC pipe of OD 24 mm. Each arm had five helical turns. The minimum flow rate at which the designed reactor can be operated to obtain a value of Dean number (De) ≥ 3 is 1 mL/min. At this flow rate the reactor offers a residence time of 55 min. However, the reactor residence time can be controlled by increasing length of the reactor by adding extra base arms. RTD for the viral inactivation reactor was performed by step injection of tracer (Bovine albumin fraction-V) as per previously established method [24]. Using the tracer data an RTD curve was plot between dimensionless concentration F(θ) and dimensionless time (θ) (Supplementary Figure S1). From the RTD curve, relative width (R_w) was calculated. Relative width is the ratio of θ at 0.5 % breakthrough to θ at 99.5 % breakthrough. R_w is used as a measure of the narrowness of the RTD curve and can attain a maximum value of 1 [24]. For the designed viral inactivation reactor, relative width of 0.69 was obtained.

2.5. Viral inactivation study for the continuous process

For the viral inactivation study, the neutralized Protein A sample was spiked with XMuLV virus culture to a final virus concentration of 5% (v/v). The spiked mAb sample was then pumped at a flow rate of 0.9 mL/

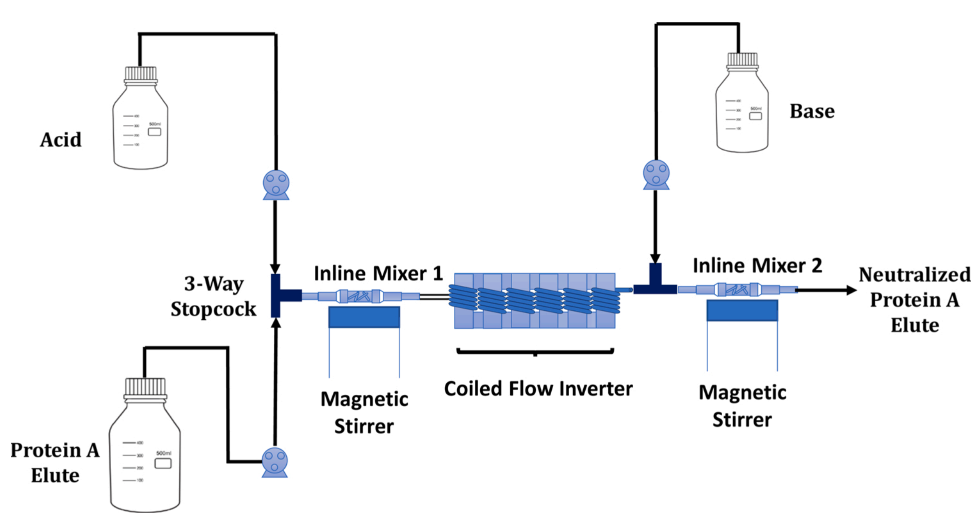

Fig. 1. Illustration of the continuous virus inactivation reactor configuration depicting the flow schematic for the viral inactivation study.

min and the acid (0.2 M HCL) was pumped at a flow rate of 0.1 mL/min (Fig. 1). A three-way stop cork joined the two inlet streams (mAb and acid) to the viral inactivation reactor. The pH of the spiked sample was decreased to 3.5 ± 0.2 in the inline mixer 1. PTFE 'T' connectors at a number of locations (5 min, 12 min and 25 min) throughout the CFIR tubing were used for withdrawal of in-process samples to understand inactivation kinetics. The mAb samples, HCl, Tris stock solutions and the sampling criteria used in the continuous configuration were all identical to those used in batch study. Two runs (duplicates) were performed using this protocol for viral inactivation assessment.

2.6. Integration of the continuous virus inactivation reactor to the continuous downstream train

Two integration strategies have been proposed in the study- strategy A and B. Details of both the strategies are provided below.

2.6.1. Integration strategy A

Protein A chromatography was performed continuously in the periodic counter current chromatography (PCC) mode on the Cadence™ BioSMB PD system. A 3 column setup of Protein A chromatography was utilized for processing the mAb feed harvest. BioSMB method was created using the BioSMB method creation tool. Time matching constrains were taken into account while selecting column bed volumes and flow rates for various chromatographic steps. The principles of counter-current mode chromatography (PCC) and its scheduling elucidated in our previous publications were followed [3,6]. Details of the continuous Protein A chromatography are given in Supplementary Table S1. All chromatography process steps were performed at a residence time of 2 min. Elution from the columns was discontinuous. With each column undergoing elution every 46 min. No peak cutting was implemented in the elution block and the entire elution block (5CV) was collected as elute. The elution profile undergoes a pH change from pH 7 of wash buffer to pH 3.5 of elution buffer. Each collected elute has an average pH value of around 4.6.

Integration of Protein A chromatography to the viral inactivation reactor is achieved with the help of a CSTR referred to as surge vessel (Fig. 2). Surge vessels are utilized for integration of unit operations, especially which function in semi-continuous and cyclic manner [29]. The surge vessel has a volume of around 250 mL. Initially, elute from 4 column runs is collected in the vessel, before starting the out flow from the vessel into the viral inactivation reactor. A mass balance equation (Eq. 3) was used to calculate the out flow rates of the stream leaving the surge vessel.

$$(Q_{in,ProteinA} - Q_{Out,VI})t_{elution,ProteinA} = (Q_{Out,VI})t_{nonelution,\ Protein\ A} \quad (3)$$

Where, $Q_{in,ProteinA}$ is the elution flow rate of the Protein A chromatography, $Q_{Out,VI}$ is the output flow rate from surge vessel into the viral inactivation reactor, $t_{elution,ProteinA}$ refers to total time for which elution occurs in the Protein A column, and $t_{nonelution,Protein\ A}$ is the time between two consecutive elutions.

Utilizing this flow rate a steady state (contents) was maintained inside the surge vessel. Sample from the surge vessel was pumped continuously into the continuous viral inactivation reactor (Fig. 2), where it was acidified by mixing of acid into the sample in the inline mixer 1, followed by low pH incubation of around 55 min in the CFIR reactor. The output from the CFIR reactor was continuously neutralized by mixing of base into the sample in the inline mixer 2. Output from the inline mixer 2 can be directly loaded onto a continuous polishing step chromatography.

2.6.2. Integration strategy B

For this integration strategy, a different method of continuous Protein A chromatography was utilized. The new method (Run B), contrary to the previous method (Run A) provided continuous elution from the Protein A chromatography and consists of a 6 column setup. The mAB harvest was initially concentrated around 4.5 times to a titer of around 8 mg/mL before being used as load. Details of the chromatography step-the time matching and buffer composition are mentioned in Supplementary Table S2.

The Protein A elute was continuously pumped into the inline mixer 1, where it was mixed with an buffer solution of varying molarity (mixture of water and 0.1 M HCl) at a constant flow rate (0.2 mL/min) adjusting the pH of the elute stream to 3.5 ± 0.2 (Fig. 3). A programmable gradient pump was used for pumping varying concentrations of acid (HCl) as a function of time. The pump was programmed to be switched on during the chromatography elution block. The acidified sample then moves through the CFIR for an incubation period of 55 min after which it is neutralized and can be loaded onto a continuous polishing step chromatography.

2.7. Analytical methods for qualitative and quantitative assessment of mAb

Analysis of mAb purity (charge variants and aggregates) was performed using analytical cation exchange HPLC (CEX-HPLC) and size exclusion HPLC (SEC-HPLC). mAb concentration was measured using analytical Protein A HPLC (PA-HPLC). Detailed methods (column details, buffer, sample loading and flow rate) for all these analytical chromatography (CEX, SEC and PA) have been laid down in our previous publications and the same were followed [3,24]. An Agilent 1260 Infinity II HPLC system was used to run all the analytical methods.

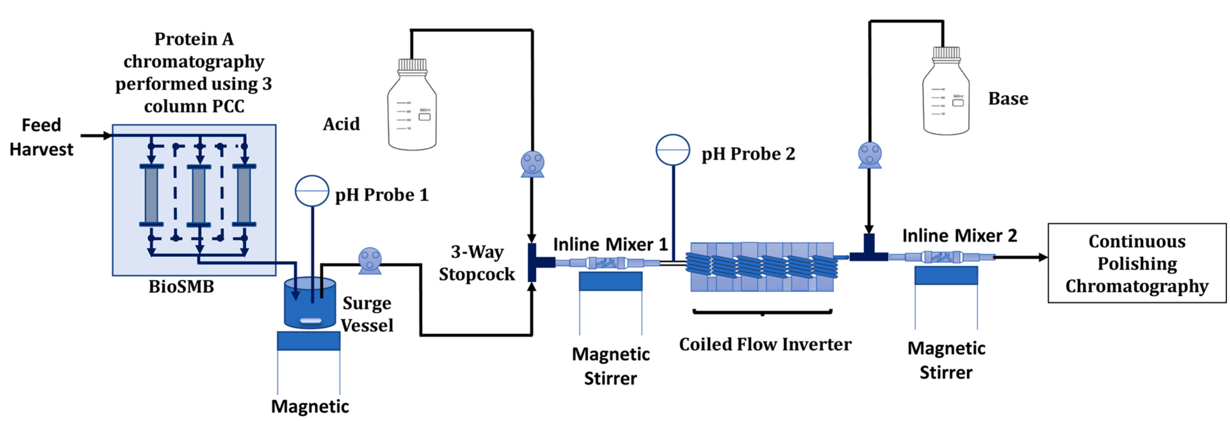

Fig. 2. Continuous downstream train showcasing integration of the continuous Protein A chromatography to continuous viral inactivation step using a surge vessel.

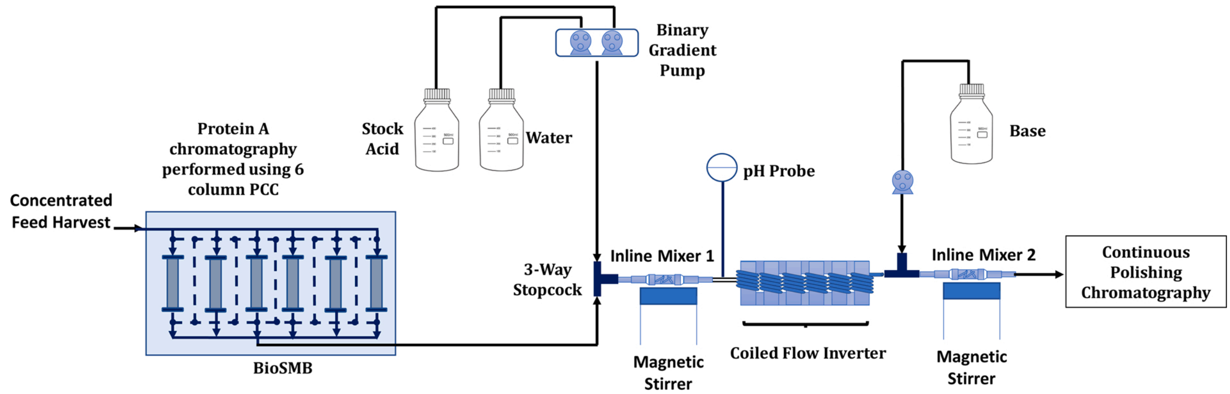

Fig. 3. Continuous downstream train showcasing integration of the continuous Protein A chromatography to continuous viral inactivation step using an inline buffer dilution setup.

3. Results and discussion

3.1. Viral inactivation assessment

The main objective of the current study was to evaluate the performance of the designed novel reactor for viral inactivation. To assess the same, low pH viral inactivation study was carried out in batch as well as the continuous reactor. Based on the finding of the study, the achieved LRVs and the kinetics of virus inactivation in the batch setup will be compared with the continuous reactor configuration.

3.1.1. Cell-based cytotoxicity and interference evaluation

Cell-based cytotoxic evaluation assay was done to find out the possible cytotoxic effects of protein that may hamper the virus induced cytopathic test. Samples that showed morphological changes >20 % in the indicator cell line (PG-4) compared to negative controls were considered as cytotoxic. The lowest dilution of protein that had no effect on the morphology of cells was found to be at 2 fold dilution.

Interference evaluation was performed to test the possible interference of mAb A on the assay test system used to titrate X-MuLV. The sample dilution that did not show difference in virus titers by $\pm$ 0.5 $\log_{10}$ compared to positive control was considered as non-interference dilution, which was again found to be 2-fold dilution. The negative control used did not show the presence of virus while the positive control used showed the presence of plaques cytopathic effect.

3.1.2. Batch low pH viral inactivation

4 cycles of continuous Protein A chromatography (Supplementary Table S1) were carried out to generate material for the viral inactivation studies. Elution from all these runs were immediately neutralized and pooled. This pooled elute was then used in two parts to carry out viral inactivation study in both batch and continuous mode. This ensured that the sample being utilized in the study was representative of the continuous chromatography elute and there was no biases from the sample end while comparing the batch and continuous process.

The batch study was carried out in a 250 mL bottle maintained at constant stirring on a magnetic stirrer. Titrated value of acid was added for acidification in the virus spiked sample. A pH probe was always dipped in the bottle so as to ensure that the desired pH (3.5 $\pm$ 0.2) was maintained during the experiment. Fig. 4A depicts the reduction in the viral titer and the corresponding LRV at different time points. It is observed that close to 4 log reduction was obtained within the first 5 min of incubation, with slight increase in reduction to around 4.3 log at the final time point of 55 min.

3.1.3. Continuous low pH viral inactivation

The proposed viral inactivation reactor (Fig. 1) consisting of two inline mixing units and a CFIR was utilized for the continuous study. The same acid and base stocks used in the batch study were utilized for the continuous runs. Contrary to the batch setup, no online pH probe was utilized to validate the acidification pH during viral spiking study due to facility limitations. In order to ensure that the desired pH value was attained when the sample enters the CFIR, a dummy run was performed with unspiked protein sample. Titrated value of acid and the mAb were mixed in the dynamic inline mixer 1 (Fig. 1). The inline dynamic mixer 1 provided vigorous initial mixing and ensured that the sample entering the CFIR had the target low pH value (3.5 $\pm$ 0.2). Fig. 4B highlights the performance of the viral inactivation study in the continuous setup. Similar to batch experiment, near to 4 log reduction is obtained in the first 5 min, followed by small increment in the LRV to around 4.12 at the final time point.

3.1.4. Comparison between batch and continuous virus inactivation

In order to understand the kinetics of the viral inactivation process a

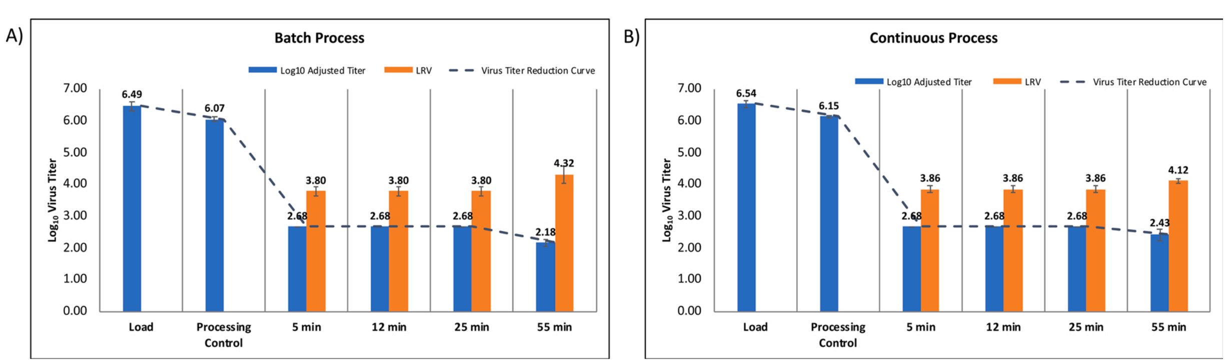

Fig. 4. Viral inactivation data (Log_{10} adjusted virus titer and the log_{10} reduction value) at different time points for A) Batch process and the B) Continuous process. A dotted line joining the average Log_{10} virus titer was plotted in both the curves (to represent inactivation kinetics).

dotted line joining the average $\log_{10}$ virus titer at each time point was plotted for both the batch and the continuous experiment (Fig. 4A and B). From the plot it was quite evident that under the set of operation conditions viral inactivation in both batch and continuous setup displayed very similar kinetics and achieved > 4 log viral reduction at the final time point. In order to statistically compare the performance of CFIR with the batch reactor, F-Test was performed on the final LRV for both the reactors. F test was done in Microsoft Excel 2016, using the data analysis tab at a significance level (α) of 0.05. From the analysis, the F calculated value was found to be less than the F critical value. This signifies that there is no statistically relevant difference in performance (variance) of the two reactors as the null hypothesis cannot be rejected.

Moreover, in both batch and continuous process a very rapid viral inactivation was obtained with around 4 log viral reduction in just 5 min. This is in agreement with a lot of previous finding where XMuLV has been shown to be rapidly inactivated within first few minutes at low pH incubation at pH value of ≤ 3.6 [8,18,30]. This can be advantageous for mAb samples that have poor stability at low pH. Despite the benefits, a rapid viral inactivation protocol is not implemented so far due to the perceived risk to patient safety associated with the inactivation step [18, 31] and so as to not increase regulatory risk [31]. Moreover, implementation of such rapid protocols requires reactor configuration that can offer a stringent control over reaction residence time. The designed continuous viral inactivation reactor due to its narrow RTD (Supplementary Figure S1), ensures that each fluid element spends near identical residence time inside the reactor, thereby easing the implementation of such rapid protocols. We strongly believe that these rapid protocols along with the designed novel reactor configuration will act as enablers for continuous processing.

The rapid in-line pH adjustment attained in the current continuous setup with the help of a dynamic inline mixer has the potential to solve one of the most significant problems associated with batch viral inactivation process - the pH adjustment step [15]. The pH adjustment step for any batch viral inactivation process at the large scale is time consuming- the titration process can take more than an hour, prone to operator error, not entirely reproducible, and potentially damaging to the product [15]. Therefore utilization of the continuous reactor should promote consistent process performance. However, due to the small scale of the experimentation, no difference in the pH consistency and mixing was observed in between the batch and the continuous setup in the current study.

Operation in the designed continuous viral inactivation reactor offers a number of other benefits apart from converting the batch process to continuous like increased reactor specific productivity (> 6x), better equipment utilization (5x), no process hold times and decreased demand of large process vessels. Since there is no requirement of shutdown, cleaning and filling, the continuous reactor provides higher reactor volume specific productivity (amount of protein produced per unit time per unit reactor volume) and equipment utilization. In the current study, the viral inactivation reactor has a productivity of 0.1 mg protein per minute per ml of reactor volume (assuming 55 min process time), whereas the batch reactor has a productivity of ~ 0.015 mg protein per minute per ml of reactor volume (assuming 55 min process time and 20 % headspace requirement in batch reactors and 4 h of cleaning and filling time required for each batch), thus resulting in a greater than 6-fold increase in productivity and 5-fold improvement in equipment utilization. Moreover, the continuous reactor offers a closed system for viral inactivation with minimal manual intervention thereby decreasing the risk of any external contamination.

A number of technology solutions like novel tubular reactor designs [12,16,18], automated multivessel system [15] and column based strategies [19,20] have been proposed in the literature for continuous viral inactivation. The viral inactivation reactor proposed in the current study appears as a promising alternative to all previous options. In comparison to the proposed novel tubular reactor called the Jig in Box (JIB), the CFIR based viral inactivation reactor has a much more simplistic design with equally narrow RTD. The multivessel system from Pall Life Sciences is semi-continuous and therefore adds discontinuity in the process decreasing its productivity. The column based strategies add chromatographic steps to the process making them less economical. Therefore the CFIR emerges as an economical and productive alternative for continuous viral inactivation. Moreover, the integration strategies established in the current study (Section 3.2) overcomes a number of perceived challenges associated with integration of tubular reactors and offers an implementable continuous integrated viral inactivation process.

mAb samples before and after the viral inactivation step for both batch and the continuous reactor were analysed for aggregate (SEC-HPLC) and charge isoform distribution (CEX-HPLC). Overlay of the analytical SEC and the CEX chromatograms showed near identical high molecular weight impurities and charge variant profile (Fig. 5A and B), thereby confirming that viral inactivation in the continuous reactor has no undesirable impact on the product quality attributes. Recovery for the continuous viral inactivation was also found to be comparable to the batch process (Table 1).

Stability of mAb during the viral inactivation step is dependent on the operating conditions (pH and residence time) and the physiological characteristics of the mAb involved in the study. Therefore operation in the optimized condition is very important for product stability while performing the viral inactivation step in batch or a continuous setup. Hence, the designed viral inactivation reactor can be readily utilized for any other mAb with optimization of the operating conditions.

3.2. *Integration of continuous viral inactivation reactor with the continuous downstream train*

After comparing the performance of the developed continuous viral inactivation reactor with the batch process, it was quite evident that the reactor offered a very promising process alternative. Despite all its benefits, the actual utilization of the reactor into an end-to-end continuous processing train becomes challenging due to number of integration constraints [15]. One of the biggest hindrance to the integration of the reactor comes from the continuous Protein A chromatography step present just before the viral inactivation step. In a typical 3 column Protein A continuous counter current chromatography process, the loading is continuous but the elution occurs in a cyclic continuous or semi-continuous manner, with each column elution occurring after a fixed time lag. Another major hurdle to smooth integration is the lack of homogeneity in the material entering the reactor. Elution of the bound protein from the chromatographic column occurs as a peak with variation in concentration and pH at different points of the elution profile. In view of this, two different integration strategies were identified- one utilizing a CSTR/surge tank and the other utilizing an in-line buffer dilution for efficient integration.

3.2.1. *Integration strategy A: integration of the continuous viral inactivation reactor using a stirred tank surge vessel*

Under this integration strategy, a stirred tank surge vessel (which acts as a CSTR) is placed between the continuous Protein A chromatography and the continuous viral inactivation reactor (Fig. 2). This surge tank helps in tackling the time lag between various column elutions and ensures homogenization of the protein concentration and pH of the sample before it enters the viral inactivation setup. The semi-continuous Protein A elute/ input stream entering into the surge vessel is converted into a continuous output stream by applying a mass balance across fluid entering and leaving the surge vessel (Equation 3). Moreover, in order to achieve efficient homogeneity (pH, conductivity and protein concentration) in the surge vessel, the volume of surge vessel plays an important role [32]. A surge vessel with working volume equivalent to 4 times the Protein A elution volume of a single run was found to be the optimum as it could efficiently dampens any fluctuation arising due to the input stream.

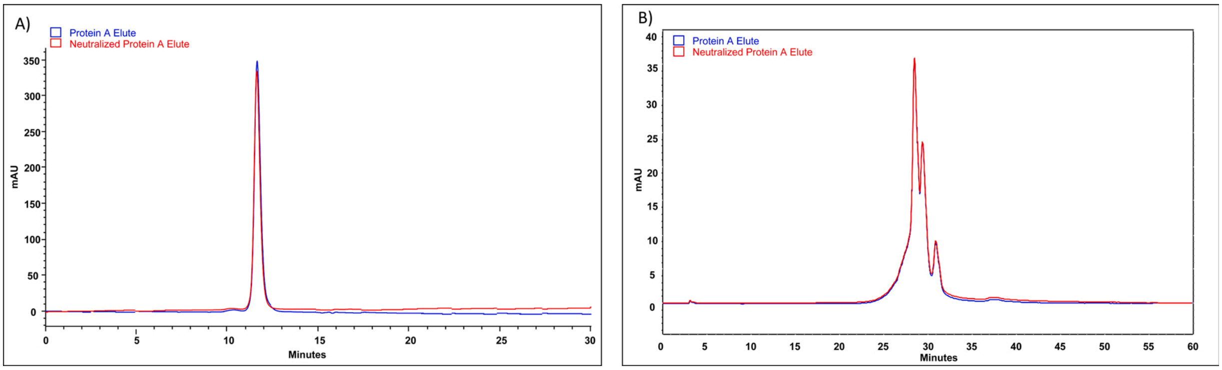

Fig. 5. Overlay of the A) Aggregate profile (SEC-HPLC) and the B) Charge variant profile (CEX-HPLC) for the mAb sample before (Protein A Elute) and after (Neutralized Protein A Elute) viral inactivation in the continuous viral inactivation reactor.

Table 1
Product quality attributes for the batch reactor and the CFIR in two integration strategies.

Sample Name	Concentration (mg/mL)	Recovery (%)	% HMWI	% Acidic	% Main	% Basic
	Batch Process					
Protein A Elute	6.7 ± 0.25	–	1.1 ± 0.04	19.3 ± 0.21	36.6 ± 0.41	44.1 ± 0.29
Neutralized Protein A Elute	5.5 ± 0.13	90.2 ± 1.35	1.1 ± 0.03	19.3 ± 0.32	36.7 ± 0.29	44.0 ± 0.30
	Integration Strategy A					
Feed Harvest	1.8 ± 0.04	–	1.2 ± 0.06	20.1 ± 0.28	36.1 ± 0.34	43.8 ± 0.31
Protein A Elute (Surge Tank)	6.7 ± 0.25	98.5 ± 1.47	1.1 ± 0.04	19.3 ± 0.21	36.6 ± 0.41	44.1 ± 0.29
Neutralized Protein A Elute	5.4 ± 0.16	91.1 ± 1.23	1.1 ± 0.05	19.1 ± 0.23	36.7 ± 0.38	44.2 ± 0.33
	Integration Strateg y B					
Feed Harvest	8.0 ± 0.15	–	1.2 ± 0.08	20.3 ± 0.32	36.4 ± 0.42	43.3 ± 0.36
Neutralized Protein A Elute	4.8 ± 0.09	88.7 ± 1.18	1.1 ± 0.06	19.0 ± 0.25	36.9 ± 0.35	44.1 ± 0.37

3.2.2. Integration strategy B: direct coupling of the continuous viral inactivation reactor by inline buffer dilution

While the attachment of a surge vessel in between the continuous Protein A chromatography and the continuous viral inactivation reactor offers a suitable solution, there is a constant debate on the use of surge vessels in the continuous processing [33]. A number of benefits associated with incorporation of surge vessels have already been realised like decreasing gradients in incoming stream, compensating for differences of flow rate, decoupling of pumps and position for monitoring and implementing process control [32–34]. Despite these benefits, it is often

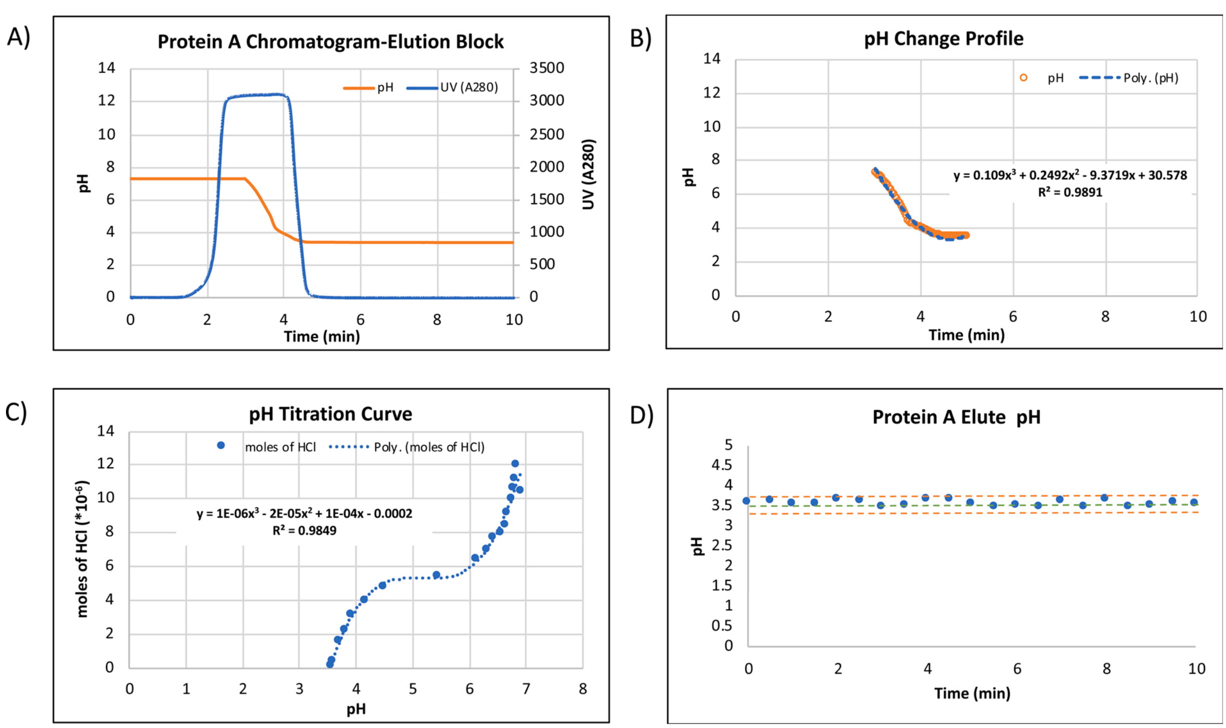

Fig. 6. A) Elution chromatogram for continuous Protein A chromatography, B) pH change as a function of time for Protein A chromatography, C) pH titration curve depicting moles of acid required for acidification at different pH values, and D) pH of the Protein A elute entering the CFIR in integration strategy B.

perceived that inclusion of surge vessels take the continuous process closer to the batch process [29,33].

Therefore an alternate integration strategy (strategy B) was implemented involving direct coupling of the viral inactivation reactor to the continuous Protein A chromatography by adding a buffer solution of varying composition (Water and 0.1 M HCl mixture) continuously to the Protein A elute stream at a fixed flow rate (Fig. 3). The acid concentration to be pumped at each point in the elution block was calculated with the help of a pH titration curve. The pH value at each point in the elution block changes as a function of time. The pH change profile during the protein A elution block was monitored with the inline pH probe of the chromatography system (Fig. 6A). The pH change curve was fit using a polynomial equation with $R^2 > 0.98$ (Fig. 6B). Moles of acid (HCl) required to decrease the pH of the sample to 3.5 ± 0.2 at different time points in the elution block was calculated using the pH titration curve (Fig. 6C). The pH titration curve was also fit using a polynomial equation with $R^2 > 0.98$. The buffer solution was mixed with the protein solution at the reactor inlet with the help of a dynamic inline mixing unit. The pH was measured using an inline pH probe (Catalogue number 18-1134-84, GE healthcare) and was found to lie within the target inactivation pH range of 3.55 ± 0.1 (Fig. 6D). The schematic of the integrated train is shown in Fig. 3.

However, in order to implement this integration scheme an important requirement that has to be fulfilled is that of the continuity in the Protein A elution stream, which cannot be attained in a regular 3 column PCC setup. For achieving continuity, the number of columns has to be increased drastically (≥ 6) with specific time matching constrains. In the current study, the input harvest concentration was increased ~4.5 folds, decreasing the loading switch time and thereby increasing the number of required columns in PCC mode to 6 along with continuous elution (Supplementary Table S2).

While the continuous stream entering in the reactor had a fixed pH value (due to buffer addition at inline mixer 1), fluctuations existed in protein concentration at different time points in the reactor. This however did not affect the virus inactivation efficiency as previous studies have already highlighted that the viral inactivation efficiency at pH 3.5 $\pm$ 0.2 remains unaffected by mAb concentration varying in the range of 5−30 mg/mL [18].

While this integration strategy eliminated the need for surge vessel, it does have its own set of drawbacks- like the need for larger number of Protein A columns in PCC mode (≥ 6) which adds additional operational complexity. Moreover, the elution stream undergoes dilution, thereby decreasing productivity of the overall setup. Hence, both the integration strategies offer their own set of pros and cons and selection of the strategy is case specific.

Both the integrated setups were run for 3 cycles and process parameters and product quality attributes were found to be consistent throughout the entire run (Table 1). Performance of both the integrated setups in terms of product recovery and product quality attributes was found to be comparable to the batch process (Table 1). The operating conditions (pH and residence time) at which the viral inactivation studies (section 3.1.3) were performed remain the same even after integration into the continuous downstream train, therefore the inactivation performance (LRV) can be safely assumed to remain unaffected.

4. Conclusions

In this study, a novel reactor configuration has been established for continuous viral inactivation in mAb manufacturing. The reactor configuration consisted of two dynamic inline mixers with the CFIR present between them. The CFIR due to its narrow residence time distribution provided the required residence time, whereas the dynamic inline mixing units provided efficient acidification and neutralization of the protein sample. Viral inactivation study for XMuLV was performed in the designed reactor and its performance was compared to traditional batch setup. While the reactor offered comparable performance in terms of virus LRV (> 4 LRV) and inactivation kinetics, it delivered significant gains over batch inactivation including 6 folds higher reactor specific productivity, 5 folds higher equipment utilization, and reduced risk of external contamination. Based on these findings, the reactor configuration has also been integrated with an existing continuous mAb downstream train. Various constraints associated with efficient integration of the reactor into the downstream train were identified and solved by designing suitable integration strategies. Two different case studies depicting the integration of the reactor to a continuous Protein-A capture chromatography under different set of operational conditions have been presented. It has been demonstrated that in either case, consistent process performance and product quality attributes were obtained.

While the current work clearly establishes the viral clearance capability of CFIR and integrates it with the continuous downstream train; future work needs to be focussed towards creation of protocols for validation of continuous viral inactivation. Creation of faster and economical methods for viral detection are also much needed to reduce the timeline and cost of viral clearance studies. Moreover, regulatory guidelines also need frequent updation in order to take new manufacturing techniques into consideration and ease the actual implementation of continuous viral inactivation.

Credit author statement

NK, N and ASR conceptualized the study. NK, N and RF performed the experiments. NK and N wrote the first draft of the paper. ASR edited the final manuscript, provided the funding and supervised the project.

Declaration of Competing Interest

The authors report no declarations of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.bej.2020.107885.

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